



MUHAMMAD RAIHAN ALIFIANDRA

Instrumentation, Control Systems & Electronics Engineer

087741174222 | raihan24trse@mahasiswa.pcr.ac.id | <https://www.linkedin.com/in/muhammad-raihan-alifiandra-2332b9417/> | <https://github.com/rezero0x00>

Pekanbaru, Riau, Indonesia

A 5th-semester Electronics Engineering Technology student at Politeknik Caltex Riau, specializing in instrumentation and control systems. I design and implement closed-loop control systems, develop PID controllers, and perform sensor calibration to ensure precise measurement in industrial and laboratory environments.

I implement these systems on Arduino and ESP32, utilizing industrial communication protocols such as Modbus RTU/RS-485, I2C, SPI, and MQTT for reliable sensor data acquisition and real-time control, with monitoring and automation supported through LabVIEW and Python.

Education

Politeknik Caltex Riau - Pekanbaru, Riau, Indonesia Sep 2024 - Aug 2028 (Expected)

Bachelor of Electronics Engineering, 3.75/4.00

- Relevant Courses: Instrumentation, PLC, Control Systems, and Electrical Circuit Analysis
- Research Assistant of IoMT & AI-Based Arrhythmia Monitoring System (data collection assistant)
- PKM – Digital Prayer Schedule Display System on Raspberry Pi, Masjid Baitul Maqdis (Electronics Technician)
- Robotics Training Instructor for P5 Program at SMAN 17 Pekanbaru

SMAN 2 BALIKPAPAN - Balikpapan, Kalimantan Timur, Indonesia Sep 2020 - Aug 2023

High School Diploma in Science, 90.00/100.00

- Documentation Staff, Mathematics Enthusiast Community
- Member, Islamic Student Organization (Rohis)

Projects

Aero Pendulum

- Designed and implemented a closed-loop PID control system to stabilize an aerodynamic pendulum using propeller-based actuation
- Tuned PID parameters (Kp, Ki, Kd) to achieve stable setpoint tracking and minimize steady-state error under disturbance conditions
- Integrated angle sensor feedback with microcontroller firmware for real-time control loop execution
- Demonstrated applied understanding of control system principles including stability analysis, transient response, and error correction

Multi-Stage Signal Conditioning Circuit

- Designed a four-stage analog signal conditioning circuit comprising amplification, filtering, attenuation, and comparator stages to convert weak, noisy sensor signals into clean, ADC-compatible output
- Calculated precision component values and selected key components, including an LM35 temperature sensor, LM358 operational amplifiers, and a TL431 voltage reference, to meet target signal range and noise reduction requirements
- Validated signal integrity at each stage through multimeter measurements, confirming amplification accuracy and noise attenuation performance

Skills

Instrumentation

Sensor Calibration, Signal Conditioning, Data Acquisition

Control System

PID Control, Fuzzy Logic, Closed-Loop Feedback, Control System Design

Embedded System

Arduino, ESP32, Raspberry Pi, UART, I2C, SPI, MQTT, Modbus RTU/RS-485

Programming

C/C++, Python, TypeScript, MySQL, SQLite

Tools & Software

LabVIEW, Multisim, MATLAB, Arduino IDE, Visual Studio Code, Github

Soft Skills

Leadership, Communication, Critical & Analytical Thinking, Teamwork

Organisational Experience

Intelligent Control by Human Politeknik Caltex Riau

Aug 2024 - Present

Research and Technology Division Member

A student organization focused on robotics, embedded systems, and intelligent control innovation.

- Conducted research on emerging technologies for workshop and training program development
- Selected components and designed wiring schematics/technical diagrams for workshop references
- Delivered technical materials and facilitated knowledge-sharing sessions for participants

Electronics Engineering Student Association, Politeknik Caltex Riau

Sep 2025 - Present

Research and Technology Division Coordinator

The official student association representing the Electronics Engineering Study Program at PCR.

- Led and coordinated division members in research activities and technical responsibilities
- Initiated and led theoretical learning sessions on foundational and applied electronics concepts
- Oversaw component selection and technical documentation for hands-on workshop implementation